

Le serveur ProFTP

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1. Installation du serveur ProFTPD.

Pour installer, on commence d'abord par un **apt update**
Puis on exécute **apt install proftpd**

1. Par défaut, l'utilisateur est **proftpd** et le groupe **nogroup**. Pour changer, on modifie le fichier de conf ici

```
# Set the user and group that the server normally runs at.
User proftpd
Group nogroup
```

2. Ils se trouvent dans **/usr/bin**

```
root@RAID5:/home/arthur# which ftpcount
/usr/bin/ftpcount
root@RAID5:/home/arthur# which ftpwho
/usr/bin/ftpwho
```

3. le fichier de configuration de proftpd est installé dans **/etc/proftpd/proftpd.conf**
4. Les pages man se trouvent généralement dans **/usr/share/man/man8/proftpd.8.gz**, dans **/usr/share/man/man1/ftpwho.1.gz** et **/usr/share/man/man1/ftpcount.1.gz**
5. On peut le retrouver dans **/var/log**

```
arthur@RAID5: ~
root@RAID5:/home/arthur# cd /var/log
root@RAID5:/var/log# ls -l
total 204
-rw-r--r-- 1 root root 11272 3 sept. 11:37 alternatives.log
drwxr-xr-x 2 root root 4096 5 sept. 15:40 apt
-rw-rw---- 1 root utmp 0 3 sept. 11:33 bttmp
-rw-r--r-- 1 root root 163074 5 sept. 15:40 dpkg.log
drwxr-xr-x 3 root root 4096 3 sept. 11:40 installer
drwxr-sr-x+ 3 root systemd-journal 4096 3 sept. 11:41 journal
-rw-rw-r-- 1 root utmp 292292 5 sept. 13:28 lastlog
drwx----- 2 root root 4096 3 sept. 11:33 private
drwxr-xr-x 2 root root 4096 5 sept. 15:40 proftpd
lrwxrwxrwx 1 root root 39 3 sept. 11:33 README -> ../../usr/share/doc/systemd/README.logs
drwxr-xr-x 3 root root 4096 3 sept. 11:36 runit
-rw-rw-r-- 1 root utmp 1536 5 sept. 13:28 wtmp
root@RAID5:/var/log#
```

6. **ServerName** donne le nom du serveur. **Port 21** fixe le port d'écoute. **User** et **Group** définissent l'utilisateur et le groupe. **MaxInstances** limite les connexions. **DefaultRoot** enferme les utilisateurs dans leur dossier. **AllowOverwrite on** autorise l'écrasement de fichiers. **Umask 022** règle les permissions des fichiers créés. **RequireValidShell off** permet aux utilisateurs sans shell valide de se connecter.

7. Avec la commande on voit que c'est ouvert sur le port 21 pour le ftp

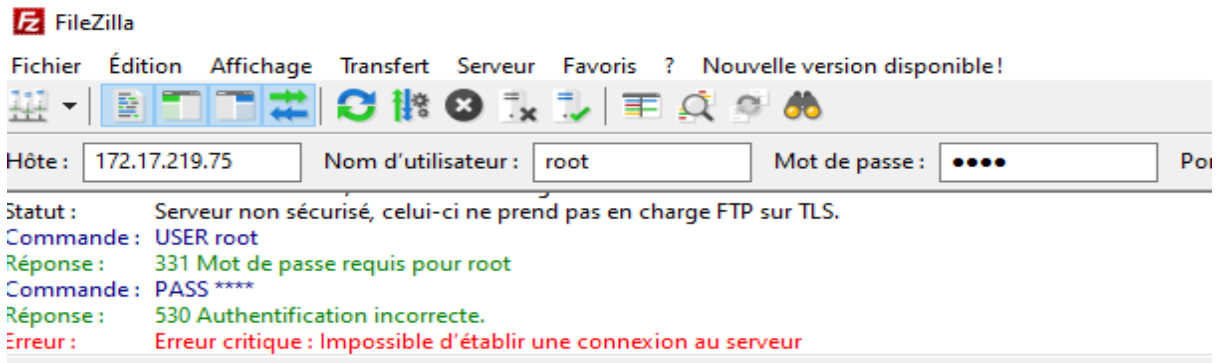
```
root@RAID5:/# nmap localhost
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-05 16:16 CEST
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000070s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
```

2. Les utilisateurs

8. La connexion en FTP est interdite
9. Il m'a placé dans /home/etudiant
10. On ne peut qu' écrire dans le répertoire perso et lire tous les autres.
11. Depuis le client FTP, l'utilisateur peut envoyer des fichiers vers son dossier personnel sur le serveur (upload) et récupérer ceux qui s'y trouvent (download). Les transferts fonctionnent tant que les permissions sur le serveur le permettent. Si un transfert échoue, c'est à cause de droits insuffisants.

3. Accès en anonyme

```
<Anonymous ~ftp>
  User ftp
  Group nogroup
#   # We want clients to be able to login with "anonymous" as well as "ftp"
  UserAlias anonymous ftp
#   # Cosmetic changes, all files belongs to ftp user
  DirFakeUser on ftp
  DirFakeGroup on ftp
#
  RequireValidShell off
#
#   # Limit the maximum number of anonymous logins
  MaxClients 10
#
#   # We want 'welcome.msg' displayed at login, and '.message' displayed
#   # in each newly chdired directory.
  DisplayLogin welcome.msg
  DisplayChdir .message
#
#   # Limit WRITE everywhere in the anonymous chroot
<Directory *>
  <Limit WRITE>
    DenyAll
  </Limit>
</Directory>
#
#   # Uncomment this if you're brave.
#   # <Directory incoming>
#   #   # Umask 022 is a good standard umask to prevent new files and dirs
#   #   # (second parm) from being group and world writable.
#   #   Umask 022 022
#   #   <Limit READ WRITE>
#   #   #     DenyAll
#   #   #     </Limit>
#   #   #     <Limit STOR>
#   #   #     # AllowAll
#   #   #     </Limit>
#   #   # </Directory>
#
</Anonymous>
```



14. On écrit ceci dans le fichier de configuration

```
# Port 21 is the standard FTP port.
Port 21

# In some cases you have to specify passive ports range to by-pass
# firewall limitations. Ephemeral ports can be used for that, but
# feel free to use a more narrow range.
PassivePorts 61000 62000
```

15. **systemctl restart proftpd**

16. Le port est bien compris entre 61000 et 62000 quand on fait une analyse de trames en se connectant.

```
[Stream index: 25]
Transmission Control Protocol, Src Port: 61747, Dst Port: 21, Seq: 51, Ack: 195, Len: 16
Source Port: 61747
```

17. **ftptop** permet de connaître l'historique des connexions au serveur tandis que **ftptop** permet de voir en temps réel qui est connecté en ce moment.

```
root@debian-12:~# ftptop
standalone FTP daemon [917], up for 21 min
 934 etudiant [ 9m29s] 9m29s idle
 939 etudiant [ 4m37s] 4m37s idle
Service class - 2 users
```

```
ftptop/1.0: Wed Sep 10 09:35:13 2025, up for 23 min
1 Total FTP Sessions: 0 downloading, 0 uploading, 1 idle
```

PID	S	USER	CLIENT	SERVER	TIME	COMMAND
939	I	etudiant	C13-02.sio.local	::ffff:172.17.219.75:21	6m31s	idle

18. Examinez les logs de transactions (téléchargements) du serveur **proftpd**

```
172.17.219.65 - PuTTY
GNU nano 7.2 /var/log/proftpd/proftpd.log
2025-09-10 08:26:12,281 debian-12 proftpd[984] debian-12.5: ProFTPD 1.3.8 (stable) (built Sat Nov 30 2024 22:32:48 UTC) standalone mode STARTUP
2025-09-10 08:38:45,974 debian-12 proftpd[984] debian-12.5: ProFTPD killed (signal 15)
2025-09-10 08:38:45,974 debian-12 proftpd[984] debian-12.5: ProFTPD 1.3.8 standalone mode SHUTDOWN
2025-09-10 08:39:04,629 debian-12 proftpd[450] debian-12.5: ProFTPD 1.3.8 (stable) (built Sat Nov 30 2024 22:32:48 UTC) standalone mode STARTUP
```

19. Pour cela, faire d'abord **man xferlog**, puis lire le fichier de log **/var/log/xferlog**.

DESCRIPTION
The xferlog file contains logging information from the FTP server daemon, proftpd(s). This file usually is found in /var/log but can be located anywhere by using a proftpd(s) configuration directive. Each server entry is composed of a single line of the following form, with all fields being separated by spaces.

current-time	transfer-time	remote-host	file-size	filename	transfer-type	special-action-flag	direction	access-mode	username	service-name	authentication-method	authenticated-user-id	completion-status
current-time	is the current local time in the form "DDD DDMM dd hh:mm:ss YYYY". Where DDD is the day of the week, DDMM is the month, dd is the day of the month, hh is the hour, mm is the minutes, ss is the seconds, and YYYY is the year.												
transfer-time	is the total time in seconds for the transfer.												
remote-host	is the remote host name.												
file-size	is the size of the transferred file in bytes.												
filename	is the name of the transferred file. If the filename contains any spaces or control characters, each such character is replaced by an underscore ('_') character.												
transfer-type	is a single character indicating the type of transfer. Can be one of: a for an ascii transfer b for a binary transfer												
special-action-flag	is one or more single character flags indicating any special action taken. Can be one or more of: C file was compressed U file was uncompressed T file was tar'd - no action was taken												
direction	is the direction of the transfer. Can be one of: o outgoing i incoming d deleted												
access-mode	is the method by which the user is logged in. Can be one of: a (anonymous) is for an anonymous guest user. r (real) is for a local authenticated user.												
username	is the local username, or if guest, the ID string given.												
service-name	is the name of the service being invoked, usually FTP.												
authentication-method	is the method of authentication used. Can be one of: 0 None 1 RFC931 Authentication												
authenticated-user-id	is the user id returned by the authentication method. A * is used if an authenticated user id is not available.												
completion-status	is a single character indicating the status of the transfer. Can be one of: c complete transfer i incomplete transfer												

20. Comparez ce que la commande ftpstats vous donne et ce qui est dans ce fichier.

```
Total Transfers from each Archive Section (By bytes)
```

Archive Section	Files Sent	Bytes Sent	---- Percent Files Sent	Of Bytes Sent
/etc/groff	1	1042	50.00	85.97
/srv/ftp	1	170	50.00	14.03

5. L'antivirus clamav

```
root@debian-12:~# systemctl stop clamav-freshclam
root@debian-12:~# freshclam
Wed Sep 10 11:09:13 2025 -> ClamAV update process started at Wed Sep 10 11:09:13 2025
Wed Sep 10 11:09:13 2025 -> daily database available for update (local version: 27758, remote version: 27759)
Current database is 1 version behind.
Downloading database patch # 27759...
Time: 0.3s, ETA: 0.0s [=====>] 2.59KiB/2.59KiB
Wed Sep 10 11:09:16 2025 -> Testing database: '/var/lib/clamav/tmp.52076caea9/clamav-ec30f9fa425358f3b8a739d3b9f65595
.tmp-daily.cld' ...
Wed Sep 10 11:09:23 2025 -> Database test passed.
Wed Sep 10 11:09:23 2025 -> daily.cld updated (version: 27759, sigs: 2076511, f-level: 90, builder: raynman)
Wed Sep 10 11:09:23 2025 -> main.cvd database is up-to-date (version: 62, sigs: 6647427, f-level: 90, builder: sigmgr
)
Wed Sep 10 11:09:23 2025 -> bytecode.cvd database is up-to-date (version: 337, sigs: 81, f-level: 90, builder: raynma
n)
ERROR: Wed Sep 10 11:09:23 2025 -> NotifyClamd: Can't find or parse configuration file /etc/clamav/clamd.conf
root@debian-12:~#
```

21. On modifie checks 24 en checks 5 dans le dossier /etc/clamav/freshclam.conf
22.

```

root@debian-12:~# clamscan -r /home
Loading:      16s, ETA:   0s [=====>]      8.71M/8.71M sigs
Compiling:    3s, ETA:   0s [=====>]      41/41 tasks

/home/etudiant/.bash_logout: OK
/home/etudiant/.profile: OK
/home/etudiant/.bashrc: OK
/home/etudiant/.bash_history: OK

----- SCAN SUMMARY -----
Known viruses: 8708239
Engine version: 1.0.9
Scanned directories: 3
Scanned files: 4
Infected files: 0
Data scanned: 0.00 MB
Data read: 0.00 MB (ratio 0.00:1)
Time: 20.541 sec (0 m 20 s)
Start Date: 2025:09:10 11:14:05
End Date:   2025:09:10 11:14:25
root@debian-12:~# █

```

23. Configurer, par exemple, un scan quotidien à 22H00 sur le home de l'utilisateur toto.

```

root@debian-12:~# crontab -e
no crontab for root - using an empty one

Select an editor. To change later, run 'select-editor'.
 1. /bin/nano      <---- easiest
 2. /usr/bin/vim.tiny

Choose 1-2 [1]: 1

# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
0 22 * * * clamscan -r /home █

```

24. Oui

```

----- SCAN SUMMARY -----
Known viruses: 8708239
Engine version: 1.0.9
Scanned directories: 3
Scanned files: 5
Infected files: 1
Data scanned: 0.00 MB
Data read: 0.00 MB (ratio 0.00:1)
Time: 21.009 sec (0 m 21 s)
Start Date: 2025:09:10 11:22:15
End Date:   2025:09:10 11:22:36
root@debian-12:~# █

```

25. avec la commande clamscan -r -R /home

```
root@debian-12:~# clamscan -r --remove /home
Loading: 17s, ETA: 0s [=====>] 8.71M/8.71M sigs
Compiling: 3s, ETA: 0s [=====>] 41/41 tasks

/home/etudiant/.bash_logout: OK
/home/etudiant/.profile: OK
/home/etudiant/.bashrc: OK
/home/etudiant/.bash_history: OK
/home/test2: Eicar-Signature FOUND
/home/test2: Removed.

----- SCAN SUMMARY -----
Known viruses: 8708239
Engine version: 1.0.9
Scanned directories: 3
Scanned files: 5
Infected files: 1
Data scanned: 0.00 MB
Data read: 0.00 MB (ratio 0.00:1)
Time: 20.497 sec (0 m 20 s)
Start Date: 2025:09:10 11:26:17
End Date: 2025:09:10 11:26:38
```

26. Intégrer Clamav au serveur Proftpd (paquet proftpd-mod-clamav).

```
#
Include /etc/proftpd/conf.d/
<IfModule mod_clamav.c>
ClamAV on
ClamServer 127.0.0.1
ClamPort 3310
</IfModule>
```

27. Mettre en local sur le poste puis essayer de transférer(eicar ne marche toujours pas)

6. Création d'utilisateurs virtuels

28. Créer deux nouveaux fichier dans /etc/proftpd, ces derniers contiendront les

```
root@debian-12:~# touch /etc/proftpd/ftpd.passwd
root@debian-12:~# touch /etc/proftpd/ftpd.group
```

29. D'abord, on crée un dossier Astérix dans le /home, ensuite on exécute les commandes.

```
root@debian-12:/etc/proftpd# ftpasswd --passwd --name=asterix --uid=1100 --file=/etc/proftpd/ftpd.passwd --home=/home/asterix --shell=/bin/false
ftpasswd: using alternate file: /etc/proftpd/ftpd.passwd
ftpasswd: --passwd: missing --gid argument: default gid set to uid
ftpasswd: creating passwd entry for user asterix

ftpasswd: /bin/false is not among the valid system shells. Use of
ftpasswd: "RequireValidShell off" may be required, and the PAM
ftpasswd: module configuration may need to be adjusted.

Password:
Re-type password:

ftpasswd: entry created
```

Affecter l'utilisateur précédent au groupe www-data

30. ftpd.passwd ⇒

```
GNU nano 7.2                                     ftpd.passwd
asterix:$5$0bY3VWVvByVo0twD$EFP6w05X2Pz1sotVySMuk9AwLxx8wUdrLwvm9YTsMf/:1100:1100:/home/asterix:/bin/false
```

ftpd.group ⇒

```
root@debian-12:/etc/proftpd# cat ftpd.group
www-data:x:33:asterix
```

7. Prise en charge des utilisateurs virtuels par ProFTPD

31. Editer le fichier /etc/shells pour ajouter /bin/false

```
root@debian-12:/etc/proftpd# echo "/bin/false" >> /etc/shells
root@debian-12:/etc/proftpd# cat /etc/shells
# /etc/shells: valid login shells
/bin/sh
/usr/bin/sh
/bin/bash
/usr/bin/bash
/bin/rbash
/usr/bin/rbash
/bin/dash
/usr/bin/dash
/bin/false
```

32. Configurer le fichier /etc/proftpd/proftpd.conf

```
# Use this to jail all users in their homes
DefaultRoot ~ www-data

AuthUserFile /etc/proftpd/ftpd.passwd
AuthGroupFile/etc/proftpd/ftpd.group
# Users require a valid shell listed in /etc/shell
```

33. Modifier le paramètre umask.

```
# Umask 022 is a good standard umask to prevent new files and dirs
# (second parm) from being group and world writable.
Umask 000 000
```

34. Faire appartenir le répertoire /var/www/html/sisr à l'utilisateur Asterix (uid 1100) et au groupe www-data.

```
root@debian-12:/etc/proftpd# chown 1100:www-data /home/asterix/
root@debian-12:/etc/proftpd# cd /home
root@debian-12:/home# ls -l
total 24
drwxr-xr-x 2      1100 www-data  4096 10 sept. 11:48 asterix
drwx----- 2 etudiant etudiant  4096 10 sept. 11:33 etudiant
drwx----- 2 root      root     16384 21 mai   2024 lost+found
root@debian-12:/home#
```